ClipSAT : Math for AI & Data Science

Unit 4.01: Conditional Probability & Bayes' Rule

Name: _____

Q1. In a dataset, the prior probability of class C_1 is PC_1 = 0.10 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.90
[ ] B: 0.95
[ ] C: 0.05
[ ] D: 1.10

Q2. In a dataset, the prior probability of class C_1 is PC_1 = 0.20 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.80
[ ] B: 0.90
[ ] C: 0.10
[ ] D: 1.20

Q3. In a dataset, the prior probability of class C_1 is PC_1 = 0.30 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.70
[ ] B: 0.85
[ ] C: 0.15
[ ] D: 1.30

Q4. In a dataset, the prior probability of class C_1 is PC_1 = 0.40 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.60
[ ] B: 0.80
[ ] C: 0.20
[ ] D: 1.40

Q5. In a dataset, the prior probability of class C_1 is PC_1 = 0.50 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.50
[ ] B: 0.75
[ ] C: 0.25
[ ] D: 1.50

Q6. In a dataset, the prior probability of class C_1 is PC_1 = 0.60 . What is the probability of the complement class Pneg C_1 ?

[ ] A: 0.40
[ ] B: 0.70
[ ] C: 0.30
[ ] D: 1.60

Q7. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.05 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.0450
[ ] B: 0.9500
[ ] C: 0.8500
[ ] D: 0.0050

Q8. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.10 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[ ] A: 0.0900

[ ] B: 1.0000

[ ] C: 0.8000

[ ] D: 0.0100